

Code:

```
#include <stdio.h> #include <stdlib.h>

struct Node { int data; struct Node *next; struct Node *prev; };

void insertStart(struct Node** head, int data) { struct Node* new_node = (struct
Node*)malloc(sizeof(struct Node)); new_node->data = data; new_node->prev = NULL;
new_node->next = *head;

if (*head != NULL) {
    (*head)->prev = new_node;
}
*head = new_node;

}

int main() { int n; if (scanf("%d", &n) != 1) return 0;

struct Node* head = NULL;
for (int i = 0; i < n; i++) {
    int val;
    scanf("%d", &val);
    insertStart(&head, val);
}

// Print forward
struct Node* curr = head;
struct Node* tail = NULL;
while (curr != NULL) {
    printf("%d", curr->data);
    if (curr->next != NULL) printf(" ");
    tail = curr;
    curr = curr->next;
}
printf("\n");

// Print backward
curr = tail;
while (curr != NULL) {
    printf("%d", curr->data);
    if (curr->prev != NULL) printf(" ");
    curr = curr->prev;
}
```

```
}  
printf("\n");  
  
return 0;  
  
}
```

Input:

```
16  
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
```

Output:

```
17 98 32 56 24 23 19 18 14 13 11 9 7 4 3 1  
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
```